# Zakaria **Abouhammadi** | Machine Learning Engineer

**Computer Science Student / Applied AI**

– Valencia, Spain

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## Professional Profile

**Machine Learning Engineer** intern at **Solver Intelligent Analytics**, currently designing and building the document generation subsystem for the Comú d'Encamp (Andorra) project, using a **LangGraph** pipeline and **python-docx** on top of a self-hosted vLLM instance.

Final-year Computer Science student specialized in **AI systems**, **advanced RAG** and **backend development**. Hands-on experience building **RAG pipelines** (HyDE, MultiQuery, rerankers, Chroma Cloud), **agents** with LangChain/LangGraph/CrewAI/AutoGen and **Transformer** and **CNN** models from scratch. Designed multi-agent debate systems, implemented a *Vision Transformer*, and developed scalable web applications. Advocate of **Scrum**, with experience as developer and Scrum Master, delivering robust and well-documented solutions.

## Experience

**Solver Intelligent Analytics** — **Valencia, Spain**

*Machine Learning Engineer, Internship contract · Hybrid · 5 months · 400 h* — *Jan 2026 – Present*

Built a conversational RAG assistant for the Comú d'Encamp (Andorra): a multilingual production chatbot integrating multiple functionalities for the municipal administration. One of the main features is the automatic generation of public procurement documents: through a guided conversational flow orchestrated with **LangGraph**, the system collects the required information from the user and produces the final Word document, covering twelve variants depending on the procedure type. The module covers the end-to-end flow: data extraction and validation, XML-level DOCX templating, and integration with **SharePoint** via **Microsoft Graph** to keep the client's templates and examples synchronized.

*Stack:* Python, LangGraph, python-docx, FastAPI, Docker, vLLM (Qwen3-30B), Microsoft Graph, SharePoint, RAG, prompt engineering.

## Projects

**yolo_v1** repo

*PyTorch, DDP, AMP, COCO 2017, ResNet18*

From-scratch implementation of **YOLOv1** (Redmon et al., 2016) trained on COCO 2017 with 80 classes. Two backbones (Darknet from scratch and pretrained ResNet18), $7 \times 7$ detection grid with 2 boxes per cell, the original five-term loss and distributed training (DDP) with mixed precision (AMP); ships 38 pytest tests and a detailed write-up of silent bugs.

**asistente-uv** (Bachelor's Thesis) private

*LangGraph, Ollama, ChromaDB, BM25, Playwright, FastAPI, React, Docker*

RAG assistant that answers questions about the University of Valencia using official information extracted from *uv.es*. Combines a local document store with real-time web search when information is missing, and grows automatically with each new query. ReAct-style agent on LangGraph with hybrid retrieval (BM25 + dense + RRF) and Playwright-based scraping.

**CodeScribe** repo

*Python, LangGraph, Ollama, Click, Rich, Pydantic, uv*

CLI tool that scans an entire project and adds documentation and comments to every source file using a local LLM via Ollama. Deterministic LangGraph pipeline: scan → classify → chunk → LLM → post-process → write back. Supports dry-run with diff, backups, interactive menu and automatic README generation.

: **chatbot — LangGraph RAG Assistant** repo
*LangGraph, BM25, Dense Embeddings, RRF, SQLite, Selenium*
Conversational QA system with hybrid retrieval fused via Reciprocal Rank Fusion, strict chunk-level citations, adaptive Selenium-based web enrichment and short/long-term memory backed by SQLite. Orchestrated in a single LangGraph instance with clearly contracted nodes. End-to-end multilingual.

: **llama_4_scratch** repo
*PyTorch, NumPy, Jupyter*
From-scratch implementation and step-by-step explanation of the key components of a LLaMA-style **decoder Transformer**: tokenizer, scaled dot-product attention and Multi-Head Latent Attention. Every operation commented with numerical examples and visualizations.

: **multiquery-hyde-rag-chroma-cloud** repo
*Python, RAG, LangChain, Chroma Cloud*
Advanced RAG pipeline with **HyDE** and **MultiQuery** to expand coverage, indexing in **Chroma Cloud**, and reranking to improve response accuracy; includes retrieval evaluation and reproducible documentation.

: **advanced-hyde-rag** repo
*Python, RAG, LangChain*
RAG implementation with HyDE and synthetic query generation, chunking and embeddings handling, comparison of retrieval strategies and latency analysis.

: **ranked-rag-pipeline** repo
*Python, Cross-Encoder, Sentence Transformers*
Retrieval chain with **rerankers** and filters, coverage/precision metrics and utilities for source traceability.

: **bm25-dense-retrieval** repo
*BM25, Embeddings, RRF*
Practical study of hybrid retrieval combining BM25 (lexical) with dense embeddings (semantic) and Reciprocal Rank Fusion. Performance comparison across query types.

: **data-chat** repo
*LangChain, LLMs, SQL*
Exploration of **text-to-SQL** and data chat patterns with agents: translate natural language questions into structured queries, execute them and return summarized answers.

: **MultiagentDebate** repo
*Python, LangChain/LangGraph*
Multi-agent debate system (scientific, economic, historical, psychological, refutation) with supervisors and automatic evaluation; orchestration with LangGraph and evidence logging.

: **VisionTransformer** repo
*PyTorch, Computer Vision*
Educational implementation of a Vision Transformer (ViT): patch embedding, multi-head attention, MLP head and training on vision datasets, with comparative analysis against CNN architectures.

: **PoetryTransformerPredict** repo
*PyTorch, NLP*
Bigram Transformer built from scratch ($\sim$9.5k embedding, 4 attention blocks) to generate Spanish poetry; training and sampling example.

: **GiveMeSomeCredit — Default Risk** repo
*PyTorch, scikit-learn, pandas, MLP, AMP*
Clean, reproducible baseline to predict serious delinquency within two years with PyTorch. Data audit, median imputation, quantile clipping, log1p transforms, RobustScaler, MLP with BatchNorm and GELU, mixed-precision training and early stopping on ROC-AUC.

: **Recommender Systems** repo
*Surprise, NumPy, Collaborative Filtering*
Collaborative filtering and content-based models, metrics (RMSE, MAP@K) and cross-validation; explanatory notebooks.

: **Brain Tumor Classification** repo
*TensorFlow, VGG19*
Brain tumor vs. non-tumor classification through fine-tuning of VGG19 and data augmentation; validation set evaluation.

: **Amazigh Alphabets Recognition** repo
*TensorFlow, Keras*
CNN network for tifinagh handwritten character recognition with $> 95\%$ accuracy; preprocessing and training pipeline.

: **Handwritten Alphabets & Digit Recognition** repo
*TensorFlow, Keras, MNIST*
CNN models for handwritten Latin letters and digits (MNIST), comparison with SVM/MLP and error analysis.

: **Housing Price Prediction** repo
*Python, scikit-learn*
Linear regression and Random Forest for housing price estimation; RMSE and $R^2$ comparison.

: **TiendaWeb (Klawz)** repo
*Java/JSP, MySQL*
Complete e-commerce with catalog, cart and payment gateway; MVC architecture, sessions and basic security.

: **PhongIllumination** repo
*C, OpenGL, OpenMP, MPI*
Implementation of the Phong illumination model in sequential, shared-memory and distributed parallel versions; speedup measurement.

: **minilang** repo
*Python, PLY*
Educational compiler for MiniLang with lexical, syntactic and semantic analysis, AST generation and test execution.

: **ISII_AdministrarHospitales** repo
*Java, Full-stack*
Hospital management system: medical records, inventory and access control; authentication and roles.

## Education

**Universitat de València** — **Valencia, Spain**
*Bachelor's Degree in Computer Science Engineering* — *Sep 2022 – Sep 2026 (expected)*
*Key subjects:* Data structures and algorithms, Web application development, Intelligent systems, Software engineering, Databases.

## Technical Skills

**Languages**: Python, C/C++, Java, JavaScript, HTML5, CSS3, SQL

**AI / ML**: PyTorch, TensorFlow, Keras, scikit-learn, Transformers, BERT/ViT, LangChain, LangGraph, CrewAI, AutoGen, BeeAI, Ollama, vLLM

**RAG / NLP**: ChromaDB (Cloud), FAISS, BM25, HyDE, MultiQuery, Reranking, Retrieval Evaluation, Prompt Engineering, Knowledge Graphs

**Backend**: FastAPI, Flask, Java EE (JSP/Servlets), REST APIs, python-docx

**Data**: PostgreSQL, MySQL, SQLite, MongoDB, pandas, NumPy

**DevOps**: Docker, Docker Compose, Git & GitHub Actions, Google Cloud, Linux

**Methodologies**: Scrum *(developer and Scrum Master)*

## Highlighted Skills

Advanced RAG (HyDE, MultiQuery, rerankers) · Vector databases (Chroma, FAISS) · Agents (LangGraph, CrewAI, AutoGen, BeeAI) · Transformers (BERT, ViT) · Convolutional networks · Risk modeling and forecasting · Decision trees · Image analysis · EDA & Feature engineering · Optimization & hyperparameters · PyTorch fundamentals · Document generation with templates

## Soft Skills

Multicultural teamwork · Trilingual communication (ES/EN/FR) · Analytical thinking · Time management · Continuous learning · Academic leadership

## Certifications & Courses

### Specializations & Programs

**Nov 2025**: **PyTorch for Deep Learning (Professional Certificate) — DeepLearning.AI**
ID: ZYEUD91L9FYS · Deep learning with PyTorch · Transfer learning for CV & NLP · Model optimization and deployment

**Oct 2025**: **Building AI Agents and Agentic Workflows (Specialization) — IBM**
ID: UROX1SUIDGWA · Design of agentic workflows, tool use, orchestration and evaluation

**Oct 2025**: **AI for Medicine Specialization — DeepLearning.AI**
ID: 6MOISRA2N5WK · Precision medicine, image analysis, risk modeling, decision trees, forecasting

**Sep 2025**: **RAG for Generative AI Applications (Specialization) — IBM**
ID: W2UQZCZMJHKV · End-to-end RAG system design and deployment

**Feb 2025**: **Deep Learning Specialization — DeepLearning.AI**
ID: EOAJ4XETDWMR · Neural networks, CNN, optimization, regularization, sequence models

**Nov 2024**: **Machine Learning — DeepLearning.AI (Andrew Ng)**
ID: 9PL5C8QDGYQ6 · Supervised, unsupervised, regularization, evaluation

**Nov 2024**: **IBM Introduction to Machine Learning — IBM**
ID: 59X2YJWXX9SS · Foundations of ML, supervised and unsupervised learning, model evaluation

### RAG & Agents

**Sep 2025**: **Advanced RAG with Vector Databases and Retrievers — IBM**
ID: TSXE4ZG7MDFQ

**Sep 2025**: **Agentic AI with LangGraph, CrewAI, AutoGen and BeeAI — IBM**
ID: AWXAVIH36K1F

**Sep 2025**: **IBM RAG and Agentic AI — IBM**
ID: X3774PPHATF6

**Jul 2025**: **Build RAG Applications: Get Started — IBM**
ID: 596YXNIVML4A

**Jul 2025**: **Fundamentals of Building AI Agents — IBM**
ID: ZGQWDFK0VO5H

**Jul 2025**: **Vector Databases for RAG: An Introduction — IBM**
ID: K46CQ74O5CKA

**Jul 2025**: **AI Enhancement with Knowledge Graphs — Mastering RAG Systems — Packt**
ID: 28DS7P6LJIE8

**Jul 2025**: **Agentic AI with LangChain and LangGraph — IBM**
ID: V82LDS06UQN4

**Jul 2025**: **Develop Generative AI Applications: Get Started — IBM**
ID: 2EPBGK0YLTIP

**Jul 2025**: **Build Multimodal Generative AI Applications — IBM**
ID: IW6XFBO3EGSN

### PyTorch & Deep Learning

**Nov 2025**: **PyTorch: Fundamentals — DeepLearning.AI**
ID: 6PJ2XY48LZZO · PyTorch basics, Tensors and Autograd

**Nov 2025**: **PyTorch: Techniques and Ecosystem Tools — DeepLearning.AI**
ID: CR0RUHDBQ5CS · Torch.Optim, PyTorch Lightning, ecosystem tools

**Nov 2025**: **PyTorch: Advanced Architectures and Deployment — DeepLearning.AI**
ID: CK4L1IG1BJ5L · Encoder–Decoder, ResNet, DenseNet, deployment

**Sep 2025**: **Building and Training Neural Networks with PyTorch — Packt**
ID: N4BCRQSYJ7D2 · CNNs

**Sep 2025**: **Foundations and Core Concepts of PyTorch — Packt**
ID: OJG0HNZ0ZQC9 · ML fundamentals

### NLP & Transformers

**Mar 2025**: **Transformer Models and BERT Model — Google Cloud**
ID: K2LDLD7XJIM6

**Feb 2025**: **Attention Mechanism — Google Cloud**
ID: LJPL5IN4LHON

### ML/DL Fundamentals

**2024–2025**: **Convolutional Neural Networks**
ID: Y4QTHRNL3HE7 · TensorFlow, CNN, facial recognition, object detection & segmentation

**2024–2025**: **Exploratory Data Analysis for Machine Learning**
ID: 8PVON5MQJ62R · EDA, feature engineering, statistical tests

**2024–2025**: **Improving Deep Neural Networks**
ID: CNCPQJS4KTFE · Regularization, optimization, hyperparameters

**2024–2025**: **Neural Networks and Deep Learning**
ID: KNJNB8WQWAJP

**2024–2025**: **Structuring Machine Learning Projects**
ID: Z3KQ6DQXD2GM

**2024–2025**: **Supervised Machine Learning: Classification**
ID: HYIO24EKA6SX

**2024–2025**: **Supervised Machine Learning: Regression**
ID: LDQ28KIIHHG7

**2024–2025**: **Unsupervised Machine Learning**
ID: 52M2R676600S

**Nov 2024**: **Supervised Machine Learning: Classification (Badge)**
Digital certificate

### Web, Cloud & Networks

**Feb 2025**: **Google AI Essentials — Google**
ID: X51IOTBS2BSE

**Feb 2025**: **Advanced Styling with Responsive Design — Univ. Michigan**
ID: VHSLGPRZGP26

**Feb 2025**: **Interactivity with JavaScript — Univ. Michigan**
ID: JODZD9YLKDBT

**Feb 2025**: **Introduction to CSS3 — Univ. Michigan**
ID: FDU5K7VCNMBP

**Feb 2025**: **Introduction to HTML5 — Univ. Michigan**
ID: B9898ISXJUTR

**Feb 2025**: **Introduction to Networking — NVIDIA**
ID: AUQMCFVGVRP0

**Feb 2025**: **AI Infrastructure and Operations Fundamentals — NVIDIA**
ID: NFNQ1HP3ODGH

## Languages

**Berber**: Native
**Arabic**: Native
**Spanish**: C1–C2 *Full professional proficiency*

**English**: B2–C1 *Full professional proficiency*
**French**: B2 *Professional working proficiency*

## Interests

Generative AI · Machine Learning · NLP · RAG · Databases · Compilers · Cloud Computing · Cybersecurity · Robotics · IoT · Video game development · Open Source · Optimization · Computer Vision · Drone photography · Hiking · Virtual/augmented reality